
KTU
NOTES

* Block diagram of a D.C power Supp

↳ Today almost every electronic d

a DC supply for its smooth operation

need to be operated within certain p

limits. This required DC voltage or DC

derived from single phase ac mains.

↳ A regulated power supply can convert

AC to a constant DC. A regulated p

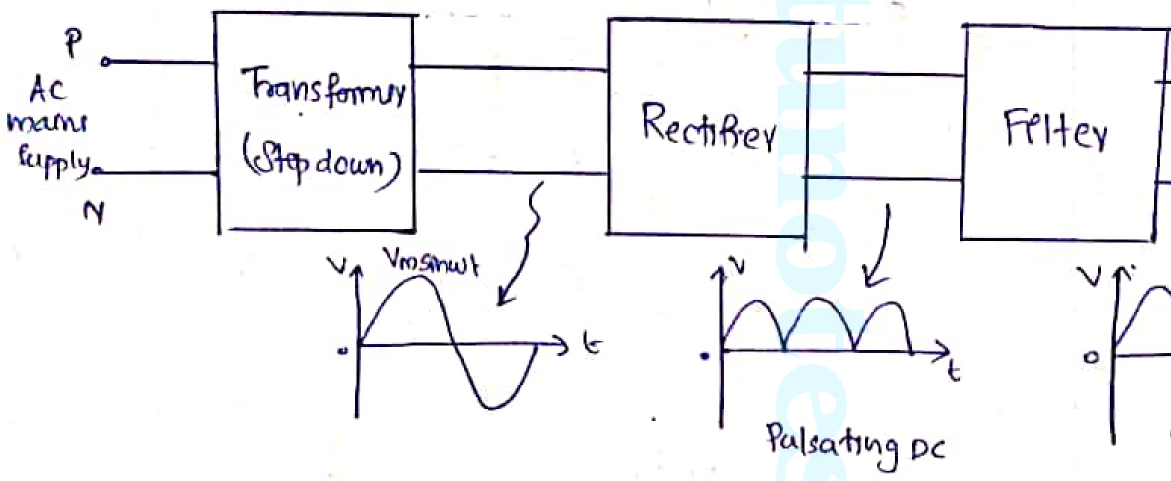
is used to ensure that the output rem

even if the input changes.

↳ Figure below shows the block diag

typical regulated DC power supply.

A DC power Supply



* Rectification

Rectifier is an electronic circuit c

diodes which carries out the rectificat

Rectification is the process of converting

voltage or current in to corresponding

quantity.

* DC filtration

The rectified voltage from the rectif

pulsating DC voltage having very high

Different types of filters are used such

filter, LC filter, π type filter etc.

Zener voltage regulator, transistor se

Fixed and variable IC regulators (IC's

* ## Rectifier

↳ Rectifiers are electronic ckt which

input ac signal in to pulsating DC

↳ Fan, heater, bulbs, airconditioner et

the help of ac signal. But there are so

which cannot be run with the help

Such as mobile phones toys, ~~etc~~ walki-ton

light torch light etc. These equipments

with the help of dc supply.

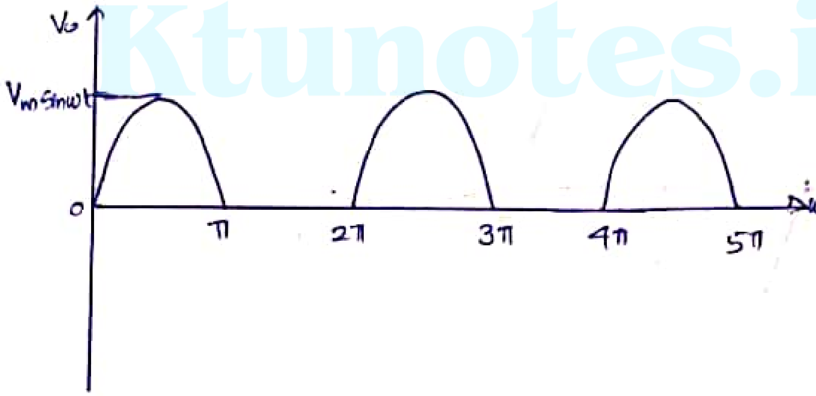
↳ As ac signals is generally availabl

equipments (electronic equipments), ac is

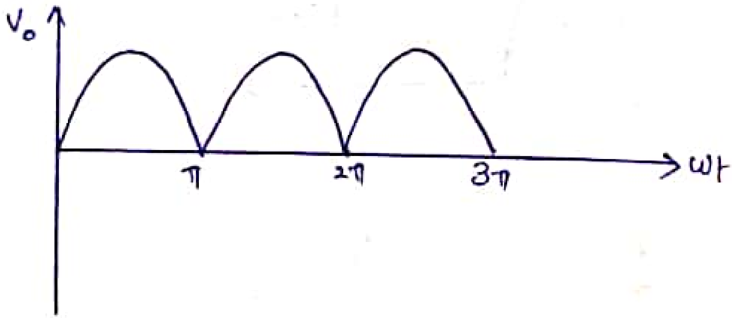
↳ Centre tap full wave rectifier

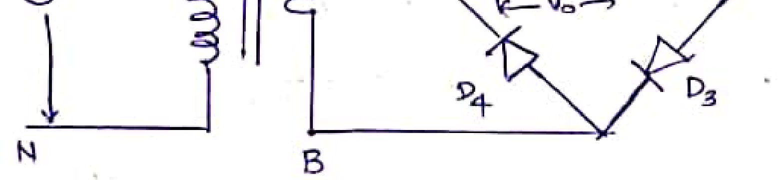
↳ Full wave bridge type rectifier

o/p of half wave rectifier

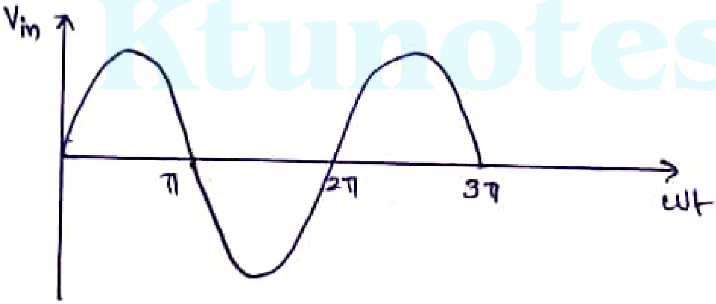


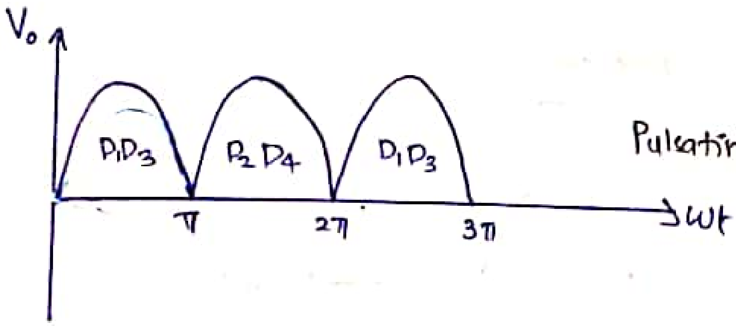
o/p of full wave rectifier





Waveforms





Pulsating D

$$V_{avg} = \frac{2V_m}{\pi} = 0.636\, V_m.$$

transformer

↳ Between other two ends of the bridg[e]

resistor R_L is connected.

Working:

↳ During the +ve half cycle of ac:

↳ During the +ve half cycle of second[ary]

the end 'A' of the secondary winding be[comes]

w.r.t. other end 'B'.

↳ Now the diodes D_1 and D_3 are forw[ard]

and hence the conventional current flo[w]

circuit across the R_L.

As a result the o/p voltage will b[e]

across R_L.

* <u>During negative half cycle of ac:</u>

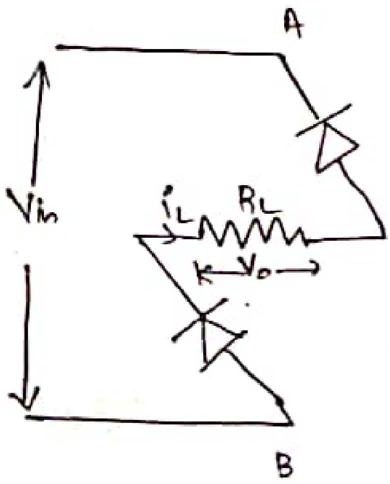
↳ During negative half cycle of second

end A becomes -ve w.r.t the other en

↳ Now the diodes D_2 and D_4 conduct c

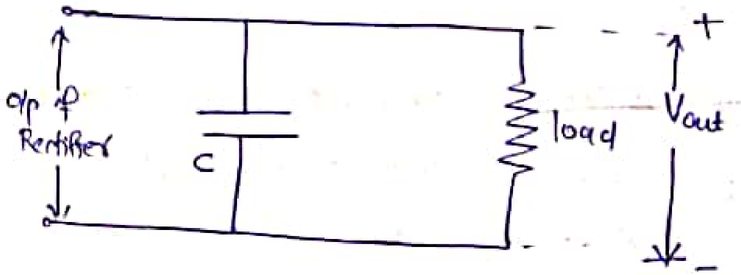
forward biasing. Hence the o/p voltage c

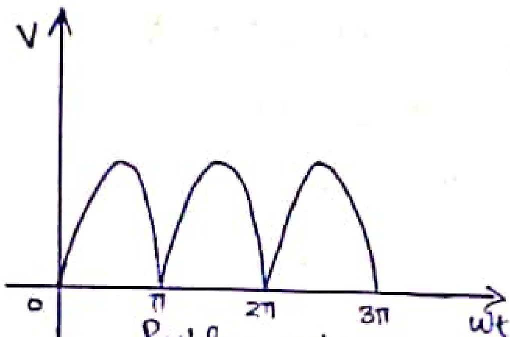
across R_L.



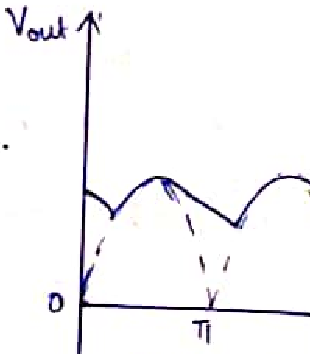
... in most applications we require ...
pure DC output voltage.

↳ Ripple content from the dc voltage is ...
connecting an electrolytic capacitor in paral...
as shown below.





Rectifier output



Rectifier o/p ...
capacitor.

↳ Zener diode behaves just like a no

purpose diode consisting of a silicon

and when biased in the forward di

is Anode positive with respect to it's

behaves just like a normal signal di

rated current.

↳ However, unlike a conventional d

blocks any flow of current through

reverse biased, that is the cathode b

positive than the Anode, as soon as

voltage reaches a pre-determined valu

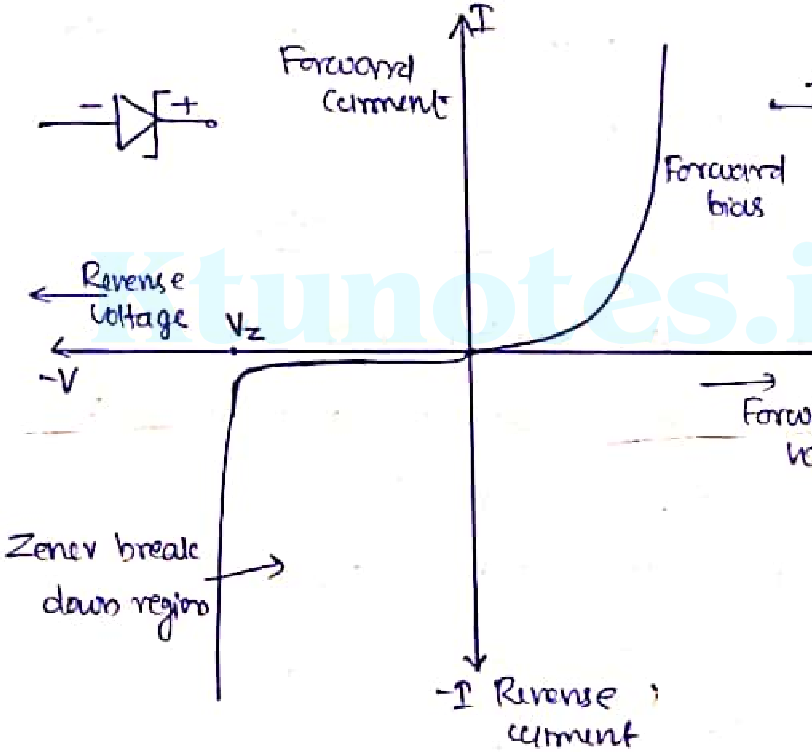
diode begins to conduct in the reverse

↳ This is happened when the voltage

break down voltage or zener voltage.

↳ A reverse biased zener diode exh

break down and the potential drop

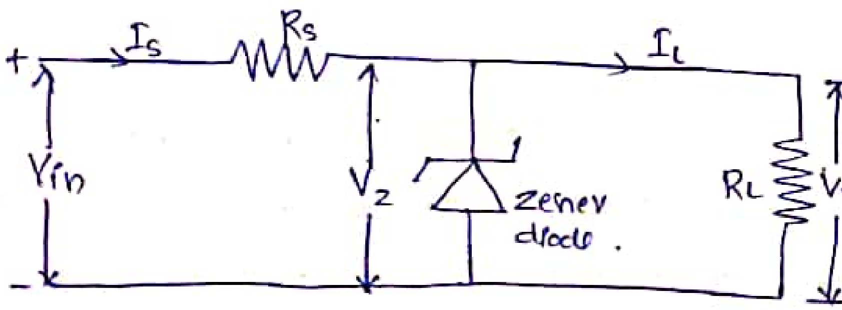


↳ From the I-V characteristics curve
can see that the zener diode has a
reverse bias characteristics of almost a
negative voltage regardless of the value

↳ This ability of the zener diode to co... can be used to regulate or stabilise a ... source against supply or load variation...

* Zener diode as a voltage regulator.

↳ Because of the ability of a Zener ... maintain a constant potential drop acro... (during reverse breakdown) they are ex... to produce a regulated voltage output. Z... stabilizer ckt is shown below.



↳ The Zener diode is connected with ... terminal connected to the positive rail of supply so it is reverse biased and wil... in its breakdown condition.

↳ Resistor R_s is selected so to limit the current flowing in the ckt.

↳ The load is connected in parallel with diode, so the voltage across R_L is alwa... same as the zener voltage. $(V_R = V_z)$

↳ Due to steep zener diode character... zener breakdown the voltage V_z doe... there by the output voltage $V_0 = V_z$ ... irrespective of variation in load curren... input voltage (V_{in}).

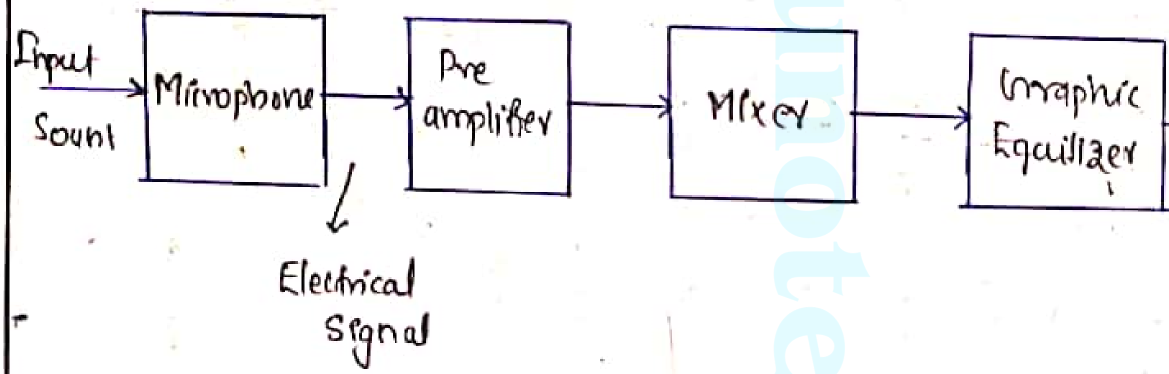
comfortably. This type of system is c
address System.

↳. It increases the loudness of a hun
musical instrument or recorded musi

* Requirements of public address syst

1. It must avoid the acoustic feedba
2. Distribute the sound intensity unifor
3. Reduce reverberations
4. It must use proper speaker orienta
5. Select proper microphones and loud spe
6. It should create a sense of directio
7. Loud speaker impedances should
   properly.

Block Diagram.



the amplitude of signal coming f
enabling for further processing.

(iii) Mixer :- The output of the microph
to the mixer stage. The mixer stage
isolate different channels from ea
before they are fed to the amplifier

(iv) Graphic Equalizer :- It is used to
balance between different frequency
within an electrical signal.

(v) Power amplifier :- Power of the el
is boosted by power amplifier so th
strong enough to drive the loud s

(vi) Loud Speaker :- It convert the
signal back in to a sound signal.

signal strength by increasing the amp
a given signal without changing its ch

↳ An RC coupled amplifier is a part
multistage amplifier wherein different
amplifiers are connected using a comb
resistor and a capacitor.

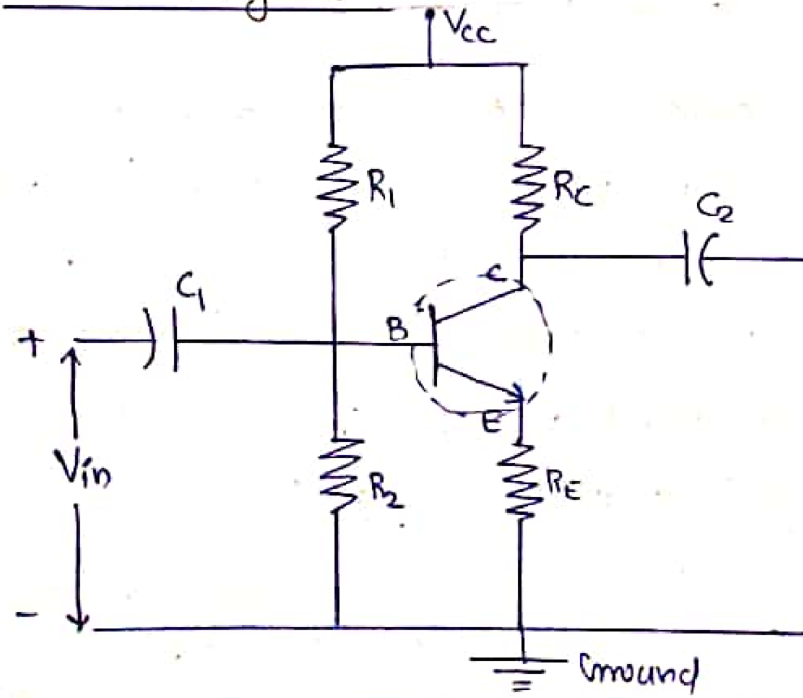
↳ Common emitter configuration of a
transistor under proper biasing condi
works as an amplifier. It will have
gain and current gain.

↳ The input signal may be a current
voltage signal or a power signal.

↳ Amplifiers are mainly used in audio
instruments communications, controllers etc

↳ Biasing can be obtained in many
Commonly used biasing is voltage d
by resistors R1 and R2 and are given t
base and collector base junctions.

* Circuit Diagram.



allow only AC voltage to the transisto

↳ R_1 and R_2 resistors are used for pro

biasing to the bipolar transistor. R_1

a biasing network which provides necess

voltage to drive the transistor in active

↳ The region b/w cut off and saturatio

known as active region. The region

bipolar transistor operation is complete

off is known as cut-off region and the r

the transistor is completely switched on is

saturation region.

↳ Resistors R_C and R_E are used f

voltage of V_{CC}. R_C → Collector resis

R_E → Emitter resis

circuit operation more stable.

Advantages of R-C coupled Amplifier.

↳ The RC coupled amplifier offers a co

over a wide frequency range.

↳ The circuite is very compact and ext

↳ It used the resistor and the capaci

are not expensive so the cost is lo

Disadvantages

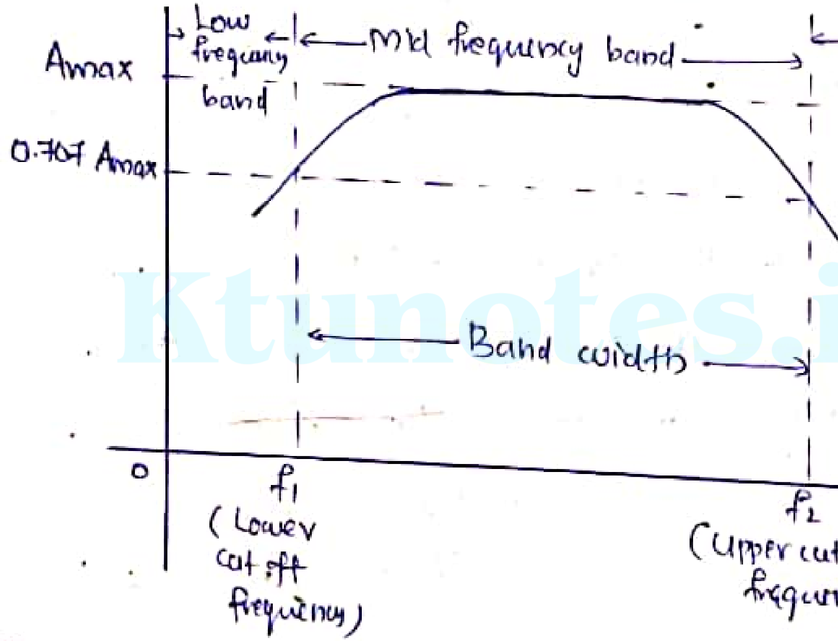
↳ The voltage and power gain are lo

of the effective load resistance.

↳ They become noisy with age.

↳ Due to poor impedance matching, p

will be low.



From the graph, it is understood t
frequency rolls off or decreases for the
below 50 Hz and for the frequencies o
~~whe~~ where as the voltage gain for
of frequencies between 50 Hz and 20
constant

Scanned w

... the frequency. At low frequencies, the ... is quite high. The reactance of input ca... and the coupling capacitor C_c are so high ... small part of the input signal is allowed ...

↳ The reactance of the emitter by pass ... is also very high during low frequen...

↳ Hence it cannot shunt the emitter re... effectively. With all these factors, the ... rolls off at low frequencies.

* At high frequencies (i.e. above 20 Hz)

Again considering the same point ... that the capacitive reactance is low at ... frequencies. So, a capacitor behaves as a ... at high frequencies. As a result of thes...

reactances. Hence the voltage gain rolls off at low frequencies.

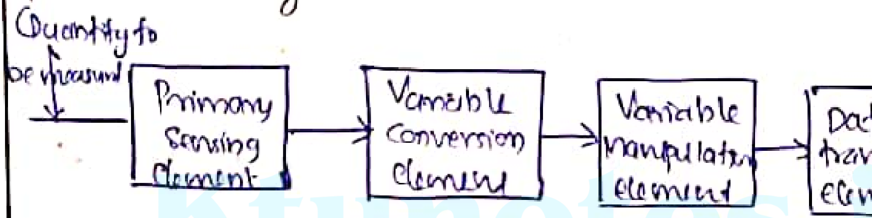
* At mid frequencies (ie, 50 Hz to 20 k

↳ The voltage gain of the capacitors is constant in this range of frequencies. frequency increases, the reactance of the C_c decreases which tends to increase

↳ But this lower capacitance reactive in loading effect of the next stage by which a reduction in gain.

↳ Due to these two factors, the gain constant.

manipulation element, data transmiss[...]
presentation system.



(i). Primary sensing element

The primary sensing element a[...]
Sensor. Basically transducers are use[d as]
sensing element. Here, the physical qu[antity is]
sensed and then converted in to anal[og]

(ii) Variable conversion element

It converts the out put of primar[y sensing]
element in to suitable from without [losing]
information. Basically these are secondar[y]

the signal in to suitable rang

(iv) Data transmission element.

Some times it is not possible to gi
read out of the quality at a perticular
In such a case, the data should tr
one place to another place through chann
known as data transmission element. T
transmission path are pneumatic pipe
cable and radio links.

(V) Data presentation or controlling ele

Finally the output is recorded or g
controller to perform action. It perfo
functions like indicating, recording or